

Conditional Diffusion Models for Exotic Option Valuation: Heavy-Tailed Path Generation and Economic Value in a P–Q Game

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This version: July 2026

Abstract

We study conditional denoising diffusion models as path generators for exotic option valuation on Chinese equity indices. Holding a conditional U-Net backbone fixed, we compare three generators: a Gaussian diffusion model (DDPM) trained with a composite finance-aware loss, the same model trained with the plain denoising loss, and a heavy-tailed Denoising Lévy Probabilistic Model (DLPM) driven by α -stable noise and trained with its standard loss. On held-out windows, the DLPM dominates both Gaussian variants on distributional fidelity—halving the Kolmogorov–Smirnov statistic and reducing Wasserstein distance and mean error severalfold—while the composite loss, not the driving noise, governs pooled tail mass at this training scale. Benchmarked further against a maximum-likelihood neural SDE and a zero-shot time-series foundation model, the DLPM offers the strongest balance of conditional shape fidelity among generators with a heavy-tail mechanism. We further propose a P–Q dynamic game in which the diffusion trader (P) quotes against a geometric Brownian motion market maker (Q) across five product families. The P-model earns robust profits on European and Asian options, while losses on lookbacks, accumulators, and deep knock-in snowballs trace directly to the Gaussian generator’s tail underestimation. Replaying the game with matched-budget generators, the DLPM converts the Gaussian model’s deep knock-in snowball losses into large stable profits with balanced positioning—distributional fidelity monetizes precisely where tails dominate.

Keywords: exotic options; structured products; diffusion models; Lévy processes; α -stable distributions; option pricing; Monte Carlo simulation.

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JEL classification: G12; G13; C45; C53; C63.

1 Introduction

Options and structured products provide contract holders with specific payoff structures based on the performance of underlying assets at predetermined times and conditions, and serve as effective tools for risk management, hedging, and portfolio optimization. As financial markets have developed, institutions have created a wide variety of exotic contracts: Asian, barrier, lookback, and ratchet options add a single feature to vanilla options, while snowball notes, phoenix notes, shark-fin options, and accumulators combine multiple path-dependent conditions with intricate payoff structures (Pesce et al., 2021).

Precisely because these products are diverse, customized, and structurally complex, accurate pricing remains a core challenge for all market participants. Buyers find it difficult to assess true value and risk; sellers face significant losses from mispricing; regulators struggle to establish unified evaluation mechanisms before risk exposures materialize. Accumulators are a cautionary example: widely sold in Hong Kong and other Asian markets in the 2000s, their path-dependent structure exposed extreme risks in volatile periods, earning the market nickname “I will kill you later.” More recently, in the 2024 Chinese A-share market, investors in snowball autocallables suffered severe principal losses when market declines triggered knock-in barriers, sparking intense public debate.

From the standpoint of practicality and versatility, simulation methods—particularly Monte Carlo simulation (Boyle, 1977)—remain the most cost-effective approach to pricing path-dependent products: simulated paths of the underlying, combined with the payoff structure, price virtually any contract. Traditional Monte Carlo, however, rests on geometric Brownian motion (GBM) dynamics with Gaussian increments, and therefore misses the heavy tails, skewness, and volatility clustering that characterize real returns (Cont, 2001). The consequences are not academic: for barrier and knock-in products, tail probabilities are the price.

Denoising diffusion models (Ho et al., 2020) approach path generation from the opposite direction: a neural network learns to progressively restore the data distribution from noise,

generating market-style paths in reverse. Their success in image and audio synthesis suggests they can capture the distributional “style” of financial paths as well. Yet the classical DDPM is itself driven by *Gaussian* noise, so whether its paths can reproduce extreme events is an open question that this paper examines directly. The recently proposed Denoising Lévy Probabilistic Model (DLPM; Shariatian et al., 2024) replaces Gaussian noise with heavy-tailed α -stable noise while retaining the computational skeleton of DDPM, providing a principled heavy-tailed alternative.

This paper makes four contributions. First, we introduce conditional denoising diffusion generators—both Gaussian and Lévy-driven—to exotic option valuation on Chinese equity indices, conditioning path generation on contract-level features (initial price, historical volatility, risk-free rate, tenor, and underlying identity). Second, we conduct a controlled ablation that isolates the role of the training objective from that of the driving noise: under one backbone and budget we compare (i) a composite finance-aware loss regularizing volatility clustering, tails, drift, quantiles, and the return spectrum; (ii) the plain denoising loss; and (iii) the standard DLPM loss with α -stable noise; two external baselines—a maximum-likelihood neural SDE and a zero-shot time-series foundation model—complete the comparison. The heavy-tailed model class delivers the largest gain in conditional distributional fidelity, while auxiliary loss engineering governs pooled tail mass at a fixed training scale. Third, we propose a P–Q dynamic game that measures the *economic* value of generated paths: the diffusion-based trader (P) quotes against a GBM Monte Carlo market maker (Q) across European, lookback, Asian, accumulator, and snowball contracts. Fourth, we provide a reproducible pipeline covering data construction, training, generation, statistical evaluation, and adversarial backtesting.

We deliberately adopt the trader’s (P-measure) perspective throughout; risk-neutral calibration of the generator into a Q-pricing model is left to future work. The remainder of the paper proceeds as follows. Section 2 reviews related work. Section 3 describes the data. Section 4 presents the Gaussian and Lévy diffusion methodologies and the three training

objectives. Section 5 reports the controlled generator ablation. Section 6 presents the P–Q game. Section 7 concludes.

2 Related Literature

2.1 Pricing structured products and exotic options

Industry practice relies on four methodologies (Deng et al., 2014): simulation, which averages discounted payoffs over generated price paths and suits strongly path-dependent products; numerical integration, when payoff and density admit closed forms; decomposition into bonds, vanilla options, and simpler exotics; and partial differential equation methods, which capture continuous state dynamics at high solution cost. The Black–Scholes model, despite its centrality for vanilla options, handles multi-layered structures poorly given its risk-neutrality assumption and single volatility parameter.

A long literature refines these tools: Monte Carlo pricing from Boyle (1977), jump-diffusion dynamics from Merton (1976), Brownian-bridge quasi-Monte Carlo constructions (Lin and Wang, 2008), ant colony optimization (Kumar et al., 2009), and tensor-network acceleration of lattice pricing (van Damme et al., 2025). Machine-learning approaches include LSTM option pricing (Pimentel et al., 2025), generative adversarial models for arbitrage-free implied volatility surfaces (Vuletić and Cont, 2025), and risk-neutral pricing via GANs (Choi et al., 2025).

2.2 Diffusion models for time series

Denoising diffusion probabilistic models were established by Ho et al. (2020), refined by the cosine schedule of Nichol and Dhariwal (2021), and accelerated by the deterministic DDIM sampler of Song et al. (2021). TimeGrad (Rasul et al., 2021) pioneered conditional diffusion for probabilistic time-series forecasting; DiffSTOCK (Daiya et al., 2024) applies relational diffusion to equity prediction; Meijer and Chen (2024) survey the area.

Diffusion models have recently become the leading generative approach for financial time series. Takahashi and Mizuno (2025) show that denoising diffusion models reproduce the stylized facts of asset returns—heavy tails, volatility clustering, and dependence structure—more faithfully than GANs and variational autoencoders, without mode collapse; subsequent work extends the framework to conditional multivariate generation with asset-specific and systematic covariates. Yet the same literature emphasizes an open problem: no generator has fully captured all stylized facts simultaneously, with tail behavior proving the most stubborn. This gap motivates our heavy-tailed model class.

Classical diffusion is Gaussian-driven. For heavy-tailed data, Shariatian et al. (2024) propose the DLPM, which replaces Gaussian increments with isotropic α -stable noise, $\alpha \in (1, 2]$ (Samorodnitsky and Taqqu, 1994). Exploiting the variance-mixing representation of stable laws, DLPM retains DDPM’s computational structure with minimal changes while markedly improving tail coverage; $\alpha = 2$ recovers the Gaussian DDPM exactly. Given the well-documented heavy tails of asset returns (Cont, 2001), DLPM is a natural generator for financial paths; to our knowledge this paper is the first to evaluate it in a derivatives-valuation setting.

2.3 Alternative model families

Two further families compete in this space. *Neural stochastic differential equations* parameterize the drift and diffusion of an SDE by neural networks, trained by likelihood or adversarial criteria (Gierjatowicz et al., 2020); imposing no-arbitrage constraints yields market models suitable for risk-neutral applications (Cohen et al., 2023). *Time-series foundation models* such as Chronos (Ansari et al., 2024) and TimesFM (Das et al., 2024) are transformers pretrained on billions of observations that produce zero-shot probabilistic forecasts; however, generic foundation models tend to underperform domain-specific architectures on financial data, and their uncertainty calibration remains an open concern. Because both families are natural alternatives to diffusion generators, our empirical study includes a conditional neural-SDE

baseline trained on identical data (Section 5.3), positioning the diffusion results against the classical SDE paradigm rather than against Gaussian Monte Carlo alone.

3 Data

3.1 Sample construction

Our underlying universe consists of broad Chinese equity indices. The CSI 1000—covering small- and mid-cap names—is a common underlying for over-the-counter options and snowball structures; we augment it with the CSI 300, CSI 500, and SSE Composite to form the pooled training universe. Daily index data from January 1, 2014 to August 1, 2025 are obtained via `akshare`. For the risk-free rate we follow OTC market-maker practice and use SHIBOR, matched to each contract’s tenor from observed maturities $M \in \{30, 60, 90, 180, 365\}$ calendar days, front-filled by date.

We employ sliding windows of 30, 90, 180, and 365 calendar days, corresponding to 1-, 3-, 6-, and 12-month risk-free tenors, sliced daily to produce historical price-path samples. The sample is split into training and out-of-sample test sets at January 1, 2024, with strict look-ahead prevention on both sides: every training window’s realized path terminates by December 3, 2023, so no training path overlaps the test period, and an embargo of roughly one month separates the last training observation from the first test window (January 2, 2024). Because windows slide daily, both sets contain overlapping observations; all reported statistics should be read accordingly, as the effective number of independent observations is smaller than the window count. The single-index CSI 1000 subsample used in the P–Q game contains 8,023 training and 827 test paths; the pooled Chinese-index sample used in the generator ablation contains 32,092 training windows.

3.2 Variable definitions and preprocessing

Let C denote calendar days and $T \subset C$ trading days. For $s, t \in C$, $CD(s, t)$ counts calendar days from s (inclusive) to t (exclusive) and $TD(s, t)$ counts trading days, so $TD(s, t) \leq CD(s, t)$. Calendar and trading times to maturity are $T_{\text{cal}} = CD(s, T)/365$ and $T_{\text{trad}} = TD(s, T)/252$. With S_t the index level on trading days, daily log returns are $r_t = \ln(S_t/S_{t-1})$, and for a window of n returns the sample mean and volatility are

$$\bar{r} = \frac{1}{n} \sum_{t=1}^n r_t, \quad \sigma_{\text{daily}} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}, \quad \sigma_{\text{ann}} = \sigma_{\text{daily}} \sqrt{252}. \quad (1)$$

Each path is transformed to scaled log-return space, $x_{0,k} = r_k/s_{\text{vol}}$ with $s_{\text{vol}} = 0.09$, so that paths are reconstructed from the initial price by exponentiating cumulative returns. Windows shorter than the maximum length of 252 trading days are zero-padded with a validity mask $M \in \{0, 1\}^{B \times L}$ ($M_{b,l} = 1$ for valid observations); all losses and statistics are computed on valid positions only:

$$L_{\text{masked}} = \frac{1}{\sum_{b,l} M_{b,l}} \sum_{b=1}^B \sum_{l=1}^L M_{b,l} (y_{b,l} - \hat{y}_{b,l})^2. \quad (2)$$

Each window carries a condition vector $c = (\tilde{S}_0, \sigma_{\text{hist}}, r_f, T_{\text{cal}}, \rho_{\text{trad}}, \text{id}_{\text{country}}, \text{id}_{\text{index}})$: five standardized numerical features (normalized initial price, trailing one-year volatility, matched risk-free rate, calendar tenor, and the ratio of trading to calendar days) plus two categorical identifiers embedded by the network (Appendix A).

4 Methodology

4.1 Gaussian conditional diffusion

The forward process of a DDPM is a Markov chain that gradually corrupts the data $x_0 \in \mathbb{R}^{B \times C \times L}$ with Gaussian noise,

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I), \quad (3)$$

under the cosine schedule β_t of Nichol and Dhariwal (2021) with $T = 500$ steps. With $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i \leq t} \alpha_i$, the marginal admits the reparameterization

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I). \quad (4)$$

The reverse process approximates $q(x_{t-1} | x_t, x_0)$ by a Gaussian transition conditioned on contract features c ,

$$p_\theta(x_{t-1} | x_t, c) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, c), \tilde{\beta}_t I), \quad (5)$$

where the conditional U-Net (Appendix A) outputs $\hat{x}_0 = f_\theta(x_t, t, c)$ and the posterior mean and variance follow from Bayes' rule for the Gaussian chain:

$$\tilde{\mu}_t(x_t, \hat{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \hat{x}_0 + \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t, \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t. \quad (6)$$

For fast generation we use the deterministic DDIM sampler (Song et al., 2021): for a subsequence $\{\tau_i\} \subset \{1, \dots, T\}$,

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}} \underbrace{\frac{x_{\tau_i} - \sqrt{1 - \bar{\alpha}_{\tau_i}} \hat{\epsilon}_{\tau_i}}{\sqrt{\bar{\alpha}_{\tau_i}}}}_{\hat{x}_0} + \sqrt{1 - \bar{\alpha}_{\tau_{i-1}} - \sigma_{\tau_i}^2} \hat{\epsilon}_{\tau_i} + \sigma_{\tau_i} z, \quad z \sim \mathcal{N}(0, I), \quad (7)$$

with $\sigma_{\tau_i} = 0$ in the fully deterministic case.

4.2 Training objective I: simple loss

The *simple* objective is the standard masked denoising regression of Eq. (2) applied to the network’s clean-sequence prediction:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| M \odot \left(f_{\theta}(x_t, t, c) - x_0 \right) \right\|_2^2 / \sum M \right]. \quad (8)$$

This is the direct analogue of the original DDPM objective and embeds no financial domain knowledge.

4.3 Training objective II: composite finance-aware loss

Financial returns exhibit stylized facts—volatility clustering, heavy tails, skewness, and characteristic spectra—that pure mean-squared error does not explicitly reward (Cont, 2001). The *composite* objective augments Eq. (8) with auxiliary regularizers computed between the predicted clean sequence $\hat{x}_0$ and the target x_0 , all masked as in Eq. (2):

$$L_{\text{jump}} = \left\| (\hat{x}_{0,i+1} - \hat{x}_{0,i}) - (x_{0,i+1} - x_{0,i}) \right\|_1 \quad (\text{relative jumps}), \quad (9)$$

$$L_{\text{vol}} = \text{SmoothL1}(\text{Std}(\hat{x}_{0,\text{win}}), \text{Std}(x_{0,\text{win}})) \quad (\text{volatility clustering}), \quad (10)$$

$$L_{\text{gvol}} = \left| \sigma_{\text{pred}} - \sigma_{\text{true}} \right| \quad (\text{global volatility}), \quad (11)$$

$$L_{\text{kurt}} = \left(K_{\text{pred}} - K_{\text{true}} \right)^2, \quad K = \mathbb{E}[z^4] - 3 \quad (\text{heavy tails}), \quad (12)$$

$$L_{\text{skew}} = \left(\mathbb{E}[z_{\text{pred}}^3] - \mathbb{E}[z_{\text{true}}^3] \right)^2 \quad (\text{skewness}), \quad (13)$$

$$L_{\text{drift}} = \left(\sum_i \Delta \hat{x}_{0,i} - \sum_i \Delta x_{0,i} \right)^2 \quad (\text{terminal drift}), \quad (14)$$

$$L_{\text{quant}} = \sum_{\tau \in \{0.01, 0.99\}} L_{\tau}(q_{\tau}(x_0), q_{\tau}(\hat{x}_0)) \quad (\text{extreme quantiles}), \quad (15)$$

$$L_{\text{spec}} = \text{SmoothL1}(\hat{P}_{\text{norm}}, P_{\text{norm}}) \quad (\text{power spectrum}), \quad (16)$$

where z denotes the standardized sequence, L_{τ} is the pinball loss of Wang et al. (2019) at quantile τ , and $P_{\text{norm}}[k] = P[k] / \max_k P[k]$ is the normalized discrete Fourier magnitude

spectrum. The extreme-quantile pair $\tau \in \{0.01, 0.99\}$ penalizes underestimation of both tails asymmetrically, sharpening the delineation of downside risk and upside potential.

Because the auxiliary terms live on heterogeneous scales, each is normalized by an exponential moving average (EMA) of its own magnitude, and all auxiliary weights are annealed linearly from zero over a warm-up phase ($s = \min(1, \text{step}/\text{warmup})$):

$$\mathcal{L}_{\text{complex}} = \mathcal{L}_{\text{simple}} + s \sum_{k \in \mathcal{K}} \lambda_k \frac{L_k}{\text{EMA}[L_k] + \varepsilon}, \quad (17)$$

with base weights $(\lambda_{\text{jump}}, \lambda_{\text{vol}}, \lambda_{\text{gvol}}, \lambda_{\text{kurt}}, \lambda_{\text{skew}}, \lambda_{\text{drift}}, \lambda_{\text{quant}}, \lambda_{\text{spec}}) = (0.5, 3.5, 8, 5, 0.5, 2, 3, 2)$.

4.4 Heavy-tailed diffusion: the DLPM

Gaussian diffusion imposes light-tailed increments on the forward chain. The DLPM of Shariatian et al. (2024) replaces them with isotropic α -stable noise, yielding a generator whose prior and reverse dynamics are heavy-tailed by construction. Let $\varepsilon^{(\alpha)} \sim \mathcal{S}\alpha\mathcal{S}(I)$ denote isotropic symmetric α -stable noise with stability index $\alpha \in (1, 2]$; $\alpha = 2$ recovers the Gaussian case. The forward chain is

$$x_t = \gamma_t x_{t-1} + \sigma_t \varepsilon_t^{(\alpha)}, \quad t = 1, \dots, T, \quad (18)$$

under a *scale-preserving* schedule: with $\bar{\gamma}_t = \prod_{i \leq t} \gamma_i$, the stability property gives the marginal

$$x_t = \bar{\gamma}_t x_0 + \bar{\sigma}_t \bar{\varepsilon}^{(\alpha)}, \quad \bar{\sigma}_t = \left(1 - \bar{\gamma}_t^\alpha\right)^{1/\alpha}, \quad (19)$$

so $\bar{\gamma}_t^\alpha + \bar{\sigma}_t^\alpha = 1$ generalizes variance preservation. We derive $\gamma_t = \alpha_t^{1/\alpha}$ from the same cosine schedule as the Gaussian model, so both models traverse comparable signal-to-noise trajectories.

Computations with stable laws become Gaussian conditionally on an auxiliary variable (Samorodnitsky and Taqqu, 1994): any isotropic α -stable vector admits the *variance-mixing*

representation

$$\varepsilon^{(\alpha)} \stackrel{d}{=} \sqrt{a} Z, \quad a \sim \mathcal{S}_{\alpha/2}^+, \quad Z \sim \mathcal{N}(0, I), \quad (20)$$

with $\mathcal{S}_{\alpha/2}^+$ the totally skewed positive $(\alpha/2)$ -stable law. Conditionally on the chain $A = (a_1, \dots, a_T)$, the forward process is Gaussian with stochastic variances and the reverse conditional $q(x_{t-1} \mid x_t, x_0, A)$ is Gaussian with

$$\Sigma_t = \sigma_t^2 a_t + \gamma_t^2 \Sigma_{t-1}, \quad \Gamma_t = 1 - \frac{\gamma_t^2 \Sigma_{t-1}}{\Sigma_t}, \quad \tilde{m}_{t-1} = \frac{x_t - \bar{\sigma}_t \Gamma_t \varepsilon_t}{\gamma_t}, \quad \tilde{\Sigma}_{t-1} = \Gamma_t \Sigma_{t-1}. \quad (21)$$

Training objective III: standard DLPM loss. The network predicts the chain noise $\varepsilon_t = (x_t - \bar{\gamma}_t x_0) / \bar{\sigma}_t$. A training sample is constructed via Eq. (20),

$$a_t \sim \mathcal{S}_{\alpha/2}^+, \quad z \sim \mathcal{N}(0, I), \quad x_t = \bar{\gamma}_t x_0 + \bar{\sigma}_t \sqrt{a_t} z, \quad \varepsilon_t = \sqrt{a_t} z, \quad (22)$$

and the *standard loss* is the L_p objective

$$\mathcal{L}_{\text{DLPM}} = \mathbb{E}_{t, x_0, a_t, z} \left[\left\| M \odot \left(\hat{\varepsilon}_\theta(x_t, t, c) - \varepsilon_t \right) \right\|_p \right]. \quad (23)$$

Because α -stable noise has infinite variance for $\alpha < 2$, the exponent must respect a moment constraint; we use the *unsquared* L_2 norm—an order-one moment of $\|\varepsilon\|_2 = \sqrt{a} \|Z\|_2$ —which is finite for all $\alpha > 1$. Shariatian et al. (2024) additionally propose a median-of-means estimator over multiple draws of a_t ; the single-draw estimator proved sufficient at our scale. For numerical stability, extreme draws of a_t are clamped at $a_{\max} = 100$, affecting only the far tail of the auxiliary variable. No financial regularizers of Section 4.3 are applied: Eq. (23) is exactly the objective prescribed by the paper.

Sampling. *Ancestral* sampling draws the auxiliary chain A and initial noise $x_T = \bar{\sigma}_T \varepsilon^{(\alpha)}$, then iterates the Gaussian posterior of Eq. (21) with $\varepsilon_t \rightarrow \hat{\varepsilon}_\theta(x_t, t, c)$. Because the variances Σ_t are driven by heavy-tailed a_t , individual reverse trajectories can take occasional large

steps—precisely the mechanism that reproduces jumps. *DLIM*, the deterministic skip-step analogue of DDIM, transports the state exactly between noise levels of a subsequence $\{\tau_i\}$:

$$x_{\tau_{i-1}} = \frac{\bar{\gamma}_{\tau_{i-1}}}{\bar{\gamma}_{\tau_i}} (x_{\tau_i} - \bar{\sigma}_{\tau_i} \hat{\epsilon}_{\tau_i}) + \bar{\sigma}_{\tau_{i-1}} \hat{\epsilon}_{\tau_i}. \quad (24)$$

5 Controlled Generator Ablation

5.1 Design

All three variants share an identical conditional U-Net backbone (17.0 million parameters; Appendix A) and an identical training budget, so the only varying factors are the driving noise and the training objective:

DDPM-simple: Gaussian diffusion, x_0 -parameterization, simple masked denoising loss, Eq. (8).

DDPM-complex: identical Gaussian diffusion, composite finance-aware loss, Eq. (17).

DLPM: α -stable diffusion with $\alpha = 1.8$, ε -parameterization, standard DLPM loss, Eq. (23).

Each model trains for 4,000 optimizer steps at batch size 64 (Adam, learning rate 10^{-4} with cosine decay, EMA weight averaging with decay 0.995) on the 32,092 pooled Chinese-index windows with $T = 500$ diffusion steps (Appendix B). Evaluation uses EMA weights and full-chain ancestral sampling on 400 held-out post-2024 test windows: for each window the model generates one path conditioned on that window’s contract features and validity mask, and per-window statistics compare generated and realized scaled log returns.

5.2 Path-level results

Table 1 reports per-window statistics; Table 2 reports statistics of the pooled return distributions.

Table 1: Per-window generation quality on held-out test windows

Metric	DDPM-simple	DDPM-complex	DLPM ($\alpha=1.8$)
Mean difference	0.0161 ± 0.0063	0.0097 ± 0.0044	0.0025 ± 0.0023
Volatility difference	0.0851 ± 0.0930	0.1128 ± 0.0657	0.1255 ± 0.1138
Kurtosis difference	3.3318 ± 4.4964	7.0425 ± 4.0269	3.3388 ± 4.8726
KS statistic	0.626 ± 0.189	0.537 ± 0.157	0.252 ± 0.103
KS p -value	0.010 ± 0.054	0.009 ± 0.038	0.293 ± 0.289
Wasserstein distance	0.0174 ± 0.0056	0.0140 ± 0.0044	0.0062 ± 0.0047
QQ R^2	0.900 ± 0.073	0.742 ± 0.114	0.915 ± 0.074

This table reports the mean $\pm$ standard deviation, across 400 held-out post-2024 test windows, of per-window discrepancies between one generated path and the realized path sharing the same contract conditions. Mean and volatility differences are absolute differences of the return mean and annualized return volatility; the kurtosis difference is the absolute difference of excess kurtosis; KS is the two-sample Kolmogorov–Smirnov statistic; QQ R^2 is the coefficient of determination of a linear fit through the paired empirical percentiles (1–99). Lower is better except the KS p -value and QQ R^2 . All models share the same U-Net backbone, training budget, and sampling protocol.

Three findings stand out. First, the DLPM with the plain paper-standard loss dominates both Gaussian variants on every distribution-shape metric: it halves the per-window KS statistic (0.252 ± 0.103 versus 0.626 ± 0.189 and 0.537 ± 0.157), is the only diffusion variant whose paths pass the KS test on average (mean p -value 0.293 ± 0.289 against ≈ 0.01 for both Gaussian variants), reduces the Wasserstein distance roughly threefold, and attains the best mean alignment and QQ R^2 . Second, the composite loss shapes the Gaussian model exactly as designed but at a price: it pushes the *pooled* excess kurtosis to 6.45—by far the closest to the real 9.48 among all trained models—yet degrades per-window shape (kurtosis overshoot of 7.0425 ± 4.0269 , QQ R^2 falling to 0.742 ± 0.114), indicating that global tail regularizers inflate extremes indiscriminately rather than matching conditional distributions. Third, at the fixed 4,000-step budget the DLPM’s pooled kurtosis (1.51) remains far below the market’s: the α -stable mechanism is necessary but not sufficient, because the bounded ε -head (a tanh output cap) and the short schedule limit how much tail mass the trained network expresses. Volatility clustering is weak across all variants, confirming that clustering must be carried by the conditional backbone rather than by the driving noise.

Table 2: Pooled return-distribution statistics

Statistic	Real	DDPM-simple	DDPM-complex	DLPM
Excess kurtosis	9.48	3.19	6.45	1.51
Skewness	0.63	−1.05	−1.32	−0.13
Annualized volatility	0.256	0.234	0.339	0.159
0.1% quantile	−0.0796	−0.0668	−0.0955	−0.0383
99.9% quantile	0.1057	0.0489	0.0880	0.0369
$ r $ autocorrelation (lag 1)	0.213	−0.124	−0.148	0.029
Pooled KS vs. real	—	0.463	0.379	0.112
Pooled Wasserstein	—	0.0129	0.0099	0.0039

This table pools daily log returns across all test windows (real column) and across the corresponding generated paths (model columns). Quantiles are of the pooled daily log-return distribution; the $|r|$ autocorrelation is the average lag-one autocorrelation of absolute returns within each window (a volatility-clustering proxy); the last two rows give the distance between the pooled generated and pooled real distributions.

5.3 Comparison with alternative model families

To position the diffusion generators against competing paradigms, we evaluate two additional baselines under the identical protocol. First, a *conditional neural SDE* (Gierjatowicz et al., 2020) trained on identical data: daily log-price increments follow $dX = \mu_\theta(h_t, c) dt + \sigma_\theta(h_t, c) dW_t$ under an Euler–Maruyama discretization, where a recurrent state h_t encodes the realized return history, the contract features c enter through an embedding, and the drift and diffusion heads are trained by exact Gaussian maximum likelihood with the same masking protocol. The model nests deep stochastic-volatility dynamics: the recurrent state lets conditional volatility respond to past returns (clustering), while innovations remain conditionally Gaussian—precisely the property the DLPM relaxes. Second, a *time-series foundation model*: the pretrained Chronos-T5-small (Ansari et al., 2024) in zero-shot mode, which receives the yearlong daily price history preceding each window’s start date as context and autoregressively samples a continuation of the window’s length; sampled levels are converted to log returns and scored identically. Unlike the other models, Chronos consumes raw price history rather than contract features, so its comparison probes what pretraining scale alone buys without domain adaptation. Its pretraining corpus predates our 2024–2025

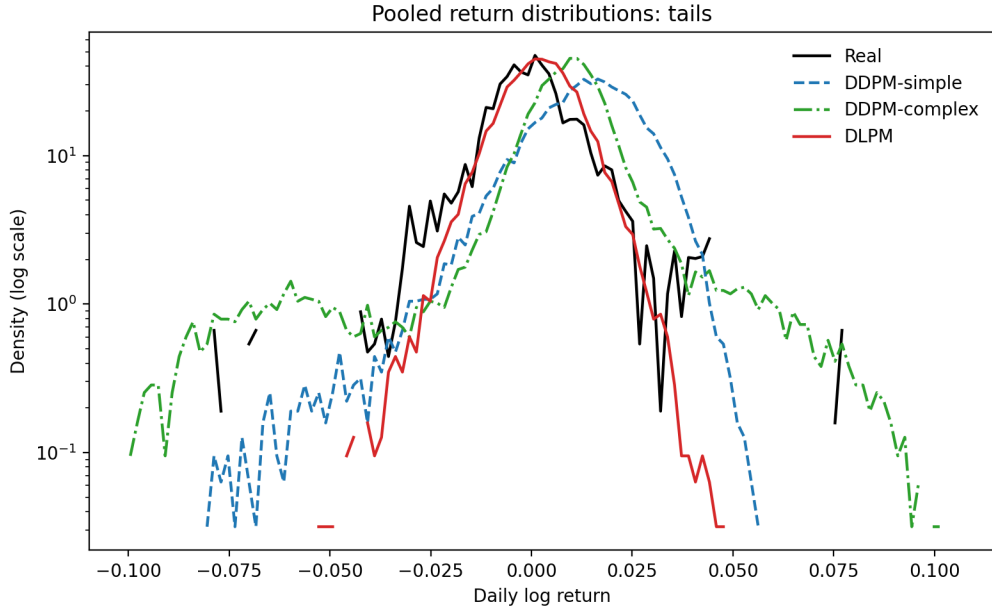


Figure 1: Pooled daily log-return densities (log scale). The figure overlays the pooled return distribution of the realized test paths (Real) with those of paths generated by the three model variants under identical conditions. Distance from the real curve in the far tails visualizes each generator’s tail fidelity.

test period and is dominated by non-financial series, so the zero-shot evaluation is free of look-ahead contamination.

Table 3 yields a nuanced cross-family picture. The neural SDE, trained by exact stepwise maximum likelihood, is the strongest marginal calibrator: it attains the lowest per-window KS statistic and Wasserstein distance, as expected for an objective that directly optimizes the conditional one-step density. Its failure is structural, not statistical: with conditionally Gaussian innovations its pooled excess kurtosis reaches only 2.90 against the market’s 9.48, and volatility clustering (0.049) falls far short of the real 0.213—no amount of training can close a gap the model class forbids. Zero-shot Chronos is the mirror image: pretraining scale buys it the most realistic *unconditional* tail mass (pooled kurtosis 12.93, extreme quantiles closest to the data), but it consumes no contract features, its per-window shape metrics trail the DLPM (KS 0.266 ± 0.080 versus 0.252 ± 0.103 ; QQ R^2 0.881 ± 0.077 versus 0.915 ± 0.074), its clustering is the weakest (0.026), and its 250-step horizon lies well beyond the 64 steps

Table 3: Alternative model families versus diffusion generators

Metric	Neural SDE	Chronos	DDPM-complex	DLPM ($\alpha=1.$
Mean difference	0.0025 ± 0.0029	0.0024 ± 0.0026	0.0097 ± 0.0044	0.0025 ± 0.002
Volatility difference	0.0862 ± 0.0808	0.0937 ± 0.0952	0.1128 ± 0.0657	0.1255 ± 0.113
Kurtosis difference	3.1633 ± 4.6684	3.4821 ± 5.0333	7.0425 ± 4.0269	3.3388 ± 4.872
KS statistic	0.202 ± 0.086	0.266 ± 0.080	0.537 ± 0.157	0.252 ± 0.103
KS p -value	0.443 ± 0.306	0.205 ± 0.254	0.009 ± 0.038	0.293 ± 0.288
Wasserstein distance	0.0050 ± 0.0037	0.0056 ± 0.0039	0.0140 ± 0.0044	0.0062 ± 0.004
QQ R^2	0.920 ± 0.065	0.881 ± 0.077	0.742 ± 0.114	0.915 ± 0.074
Pooled excess kurtosis (real: 9.48)	2.90	12.93	6.45	1.51
Pooled KS vs. real	0.031	0.172	0.379	0.112
$ r $ autocorrelation (real: 0.213)	0.049	0.026	-0.148	0.029

The neural SDE parameterizes drift and diffusion of daily log-price increments by neural networks with a recurrent state and is trained by Gaussian maximum likelihood on the same training windows, conditions, and masks as the diffusion models. Chronos-T5-small (Ansari et al., 2024) is evaluated zero-shot: for each test window it receives the preceding (up to) 252 trading-day price history of the same index as context and samples one continuation of the window’s length. Evaluation follows the protocol of Table 1 on the same 400 test windows. Definitions as in Tables 1 and 2.

the model is optimized for. The DLPM sits between the two: within striking distance of the neural SDE on every calibration metric while being the only conditional generator whose noise mechanism can, in principle, express heavy tails—making longer training, rather than architectural change, the natural route to combining both strengths.

5.4 Accelerated sampling

Replacing the 500-step ancestral chain by 50-step deterministic DLIM (Eq. 24) reduces sampling time by a factor of roughly 10 and leaves the calibration metrics essentially unchanged—the pooled KS statistic even improves slightly to 0.102. The caveat is at the extremes: with only 50 large steps, occasional windows saturate the sampler’s stability bounds, inflating the pooled excess kurtosis to 168.39. We therefore recommend DLIM for calibration-sensitive workloads (fast valuation loops such as the game of Section 6) and the full ancestral chain for tail-sensitive risk applications.

5.5 Static validation of the production P-model

Independently of the controlled ablation, the production DDPM-complex P-model used in the P–Q game (trained to full convergence on the CSI 1000 subsample) was validated statically on its 827-window test set (Table 4).

Table 4: Static validation of the production P-model (CSI 1000)

	Mean	Std
Mean difference	0.0053	0.0037
Volatility difference	0.1137	0.0922
Kurtosis difference	3.559	3.517
KS statistic	0.199	0.094
KS p -value	0.2087	0.2407
Wasserstein distance	0.00836	0.00505
QQ R^2	0.889	0.0518

Definitions as in Table 1, computed on the 827 held-out CSI 1000 windows used in the P–Q game of Section 6.

The production model excels at the core statistics: the mean difference is 0.0053, the Wasserstein distance is small, and the average KS p -value of 0.2087 means equality of distributions cannot be rejected at the 5% level; the QQ R^2 of 0.889 confirms strong quantile correspondence. The single glaring weakness is the kurtosis difference of 3.56—by far the largest discrepancy: Gaussian-diffusion tails remain “thinner” than the market’s even though the training loss explicitly penalizes kurtosis mismatch. This diagnosis is what motivates the heavy-tailed DLPM studied in the controlled ablation above, whose noise mechanism removes the structural obstacle even though fully expressing market-level kurtosis requires training beyond the ablation budget.

6 The Trader–Market-Maker Dynamic Quoting Game

As a P-measure model, the diffusion generator naturally invites expectations about out-of-sample trading performance. We therefore pair the P-model against a risk-neutral Q-model

acting as counterparty in backtests over the 827 CSI 1000 test windows.

6.1 Game framework

The Q side is a GBM Monte Carlo pricer. This choice serves two purposes: P and Q receive identical inputs, enabling clean head-to-head comparisons, and GBM Monte Carlo remains the canonical risk-neutral pricing approach. To mitigate the limitations of the baseline Q pricer, we evaluate the market maker under varying profit targets (a *greediness* parameter g): when selling, Q quotes $(1 + g)$ times its risk-neutral price; when buying, it accepts $(1 - g)$ times that price, giving it margin to profit after hedging even under model error. A trade executes only when the relative P–Q price gap exceeds 10%, a threshold intended to cover transaction costs and filter weak signals. For snowball notes, whose value is dominated by coupon terms, greediness is instead an absolute spread Δ on the note value. Because test windows slide daily, consecutive trades reference largely overlapping price paths; cumulative P&L therefore aggregates dependent positions, and Sharpe ratios computed from these trades overstate risk-adjusted performance relative to independent bets.

6.2 Product set

We consider five contract families. *European options* ($K = S_0$) evaluate accuracy on terminal prices. *Fixed-strike lookback calls* settle on the path maximum, $\text{Payoff} = \max(M_T - K, 0)$ with $M_T = \max_{t \in \mathcal{T}} S_t$, testing extreme-value fidelity. *Arithmetic-average Asian calls* settle on $A_T = |\mathcal{T}|^{-1} \sum_{t \in \mathcal{T}} S_t$, testing path-average fidelity. *Accumulators* specify a discount level $K_d = d S_0$ and a knock-out barrier $B S_0$: each day the buyer purchases q_t units at K_d ,

$$q_t = \begin{cases} 2, & S_t < K_d, \\ 1, & S_t \geq K_d, \end{cases} \quad CF_t = q_t (S_t - K_d), \quad (25)$$

and the contract knocks out once $S_t \geq B S_0$ —probing day-by-day path accuracy. *Snowball notes* combine a knock-out barrier (observed every five trading days and at expiry) with a knock-in barrier (observed daily): touching KO autocalls principal plus a high coupon; breaching KI without subsequent KO exposes principal to equity-like downside; staying between barriers pays a preset coupon. Snowball value is highly sensitive to the full path distribution—tails (KI), upside (KO), and volatility clustering through barrier-touch probabilities.

6.3 European options

Table 5: P–Q game: European call, $K = S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	177,805	153,971	135,130	115,012	94,841
Trades executed	808	786	761	737	696
Long positions	346	336	326	311	296
Short positions	462	450	435	426	400
Win rate	66.96%	65.39%	64.78%	63.77%	63.07%
Annualized Sharpe ratio	6.81	6.07	5.50	4.80	4.23

Cumulative profit and loss of the P-model (diffusion trader) against the Q-model (GBM Monte Carlo market maker) over 827 CSI 1000 test windows. The market maker quotes $(1 + g)$ times its risk-neutral price when selling and $(1 - g)$ when buying; trades execute only when the relative P–Q price gap exceeds 10%.

Even at $g = 40\%$ the P-model achieves substantial cumulative profits with a high win rate: the generator models terminal distributions accurately enough to exploit large P–Q gaps for standard European payoffs.

6.4 Lookback options

With greediness up to 20% the P-model remains profitably tradable, but because the Gaussian generator under-captures path maxima, cumulative P&L decays quickly as g rises.

Table 6: P–Q game: fixed-strike lookback call, $K = S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	125,622	86,984	40,112	−10,061	−51,036
Trades executed	787	757	710	664	620
Long positions	306	290	259	233	209
Short positions	481	467	451	431	411
Win rate	62.64%	59.05%	53.80%	48.34%	43.55%

Setup as in Table 5; the payoff references the path maximum over the contract horizon.

6.5 Asian options

Table 7: P–Q game: arithmetic-average Asian call, $K = S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	111,966	100,887	89,757	79,356	71,608
Trades executed	796	734	687	654	621
Long positions	498	461	438	428	418
Short positions	298	273	249	226	203
Win rate	59.42%	59.95%	58.52%	56.88%	55.39%

Setup as in Table 5; the payoff references the arithmetic average price over the contract horizon.

Profits remain substantial and win rates stable through $g = 40\%$: the generator models path averages with sufficient accuracy to exploit P–Q discrepancies for Asian payoffs.

6.6 Accumulators

Tables 8–11 report the four accumulator variants with discount levels $d \in \{90\%, 80\%\}$ and knock-out barriers $B \in \{120\%, 150\%\}$.

Across accumulator variants, long/short imbalances reveal which side holds the positional edge. The worst performer combines $d = 80\%$ with KO at 150%: rare knock-outs (hard to autocall) with the deepest investor concession. The mix is deceptively attractive—apparently wide profit room—but typically leads to long time-in-trade, during which distant price

Table 8: P-Q game: accumulator, $d = 90\%$, KO at $150\% S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	8,942,537	6,430,651	1,176,207	-1,061,260	-2,589,155
Trades executed	781	727	680	644	587
Long positions	385	349	316	301	279
Short positions	396	378	364	343	308
Win rate	60.82%	55.16%	48.53%	43.01%	38.33%
Annualized Sharpe ratio	4.19	3.29	0.76	-0.73	-1.91

Setup as in Table 5. The buyer purchases one unit daily at the discount price $K_d = d S_0$ (two units when $S_t < K_d$); the contract knocks out once $S_t \geq B S_0$.

Table 9: P-Q game: accumulator, $d = 90\%$, KO at $120\% S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	5,493,222	3,183,898	1,397,908	-259,511	-1,165,121
Trades executed	790	713	680	636	482
Long positions	269	256	243	212	77
Short positions	521	457	437	424	405
Win rate	65.06%	57.08%	50.29%	46.70%	46.06%
Annualized Sharpe ratio	4.71	2.98	1.39	-0.27	-1.46

moves erase early gains and “kill you later.” Lower discounts and higher KO levels make accumulators more misleading for buyers, consistent with the product’s reputation.

6.7 Snowball notes

Tables 12–16 report five snowball variants with KO barriers at 105% or 110% of S_0 , KI barriers at 90% or 80% , and annual coupons of 12% or 15% .

The product placing the highest demand on path realism yields the most nuanced picture. Because the Gaussian generator under-learns extreme events, it underestimates deep drawdowns and—under that belief—over-sells snowball notes. With KI at 90% of S_0 the risk is partly masked by mild terms and the model manages exposures reasonably. With the deeper KI at 80% , the model incorrectly infers “more room,” shifts entirely to the short side

Table 10: P–Q game: accumulator, $d = 80\%$, KO at $120\% S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	7,451,849	3,244,019	−146,262	−1,843,920	−4,089,229
Trades executed	757	576	365	274	253
Long positions	247	97	0	0	0
Short positions	510	479	365	274	253
Win rate	65.26%	58.16%	50.96%	42.34%	33.60%
Annualized Sharpe ratio	5.92	3.16	−0.19	−2.89	−6.64

Table 11: P–Q game: accumulator, $d = 80\%$, KO at $150\% S_0$

Greediness g	0	10%	20%	30%	40%
Cumulative P&L	7,093,877	373,014	−2,320,840	−5,388,933	−6,495,338
Trades executed	706	588	479	298	220
Long positions	351	302	237	87	29
Short positions	355	286	242	211	191
Win rate	62.46%	48.98%	38.83%	26.17%	14.55%
Annualized Sharpe ratio	4.10	0.31	−2.32	−7.48	−12.58

(zero long positions in Table 16), and realized paths show that a single severe downturn flips many trades from profit to large loss. *Tail underestimation is the primary failure mode of the Gaussian P-model on snowballs*—exactly the deficiency that the DLPM corrects at the path-distribution level in Table 2.

6.8 The game under matched budgets: DLPM versus Gaussian diffusion

The preceding tables evaluate the production Gaussian model. To test whether the DLPM’s distributional fidelity converts into economic value, we replay the game with the two 4,000-step generators of the controlled ablation as alternative P-models against the same GBM market maker (Table 17).

Two patterns stand out. First, the positioning structure exposes calibration. The Gaussian model at this budget takes an almost uniformly long stance—827 of 827 trades

Table 12: P–Q game: snowball, KO 105%, KI 90%, coupon 15%

Spread Δ	0	0.5%	1%	1.5%	2%
Cumulative P&L	11,288,309	8,833,011	6,975,398	5,130,497	3,396,778
Trades executed	697	679	655	639	623
Long positions	77	71	66	59	58
Short positions	620	608	589	580	565
Win rate	55.67%	54.93%	54.66%	54.15%	54.41%
Annualized Sharpe ratio	2.69	2.15	1.76	1.33	0.90

Cumulative P&L per 1 million notional. The market maker’s greediness is an absolute spread Δ on the note value; KO is observed every five trading days and at expiry, KI daily.

Table 13: P–Q game: snowball, KO 110%, KI 90%, coupon 15%

Spread Δ	0	0.5%	1%	1.5%	2%
Cumulative P&L	9,396,558	7,090,688	4,805,815	2,982,048	1,200,892
Trades executed	804	791	777	769	763
Long positions	36	25	13	8	6
Short positions	768	766	764	761	757
Win rate	42.16%	40.58%	39.12%	37.97%	37.75%
Annualized Sharpe ratio	1.83	1.39	0.95	0.60	0.24

long on lookback and Asian contracts—with win rates near 35%: its systematically inflated valuations happened to align with the 2024–2025 CSI 1000 rally, so raw P&L remains positive on vanilla-style payoffs, but the profits are directional luck rather than pricing skill. The DLPM trades both sides (for example, 359 long versus 399 short on European calls), sustains win rates of 48–59% across products, and its P&L decays gracefully with the market maker’s spread—the signature of genuinely calibrated valuations.

Second, on the deep knock-in snowball—the product family where direction cannot rescue a mispriced tail—the ranking is unambiguous. The Gaussian model loses 1.1–2.3 million per million notional across spreads, mostly short, reproducing the production model’s failure mode of over-selling deep-KI notes. The DLPM earns above 8 million at every spread level with a 59% win rate and a predominantly long book: its heavy-tailed conditional distribution prices knock-in risk on the correct side of the GBM quote. The accumulator tells the same

Table 14: P–Q game: snowball, KO 105%, KI 90%, coupon 12%

Spread Δ	0	0.5%	1%	1.5%	2%
Cumulative P&L	10,258,564	8,061,852	5,806,006	3,906,674	2,150,148
Trades executed	704	666	641	620	601
Long positions	103	79	71	66	59
Short positions	601	587	570	554	542
Win rate	55.82%	54.65%	54.29%	54.19%	53.74%
Annualized Sharpe ratio	2.78	2.28	1.70	1.18	0.67

Table 15: P–Q game: snowball, KO 110%, KI 90%, coupon 12%

Spread Δ	0	0.5%	1%	1.5%	2%
Cumulative P&L	8,271,322	6,099,404	4,041,648	2,077,619	126,650
Trades executed	808	787	762	744	725
Long positions	45	38	30	20	9
Short positions	763	749	732	724	716
Win rate	40.59%	39.90%	38.85%	37.90%	37.38%
Annualized Sharpe ratio	1.80	1.35	0.91	0.47	0.03

story in miniature: the DLPM remains profitable at every greediness level while the Gaussian model’s win rate collapses from 49% to 35%. Economic value thus tracks the statistical diagnosis of Section 5: conditional shape fidelity, not pooled tail mass alone, is what monetizes in adversarial quoting.

7 Conclusion

This paper is, to our knowledge, among the first to bring denoising diffusion generators to exotic option valuation and to evaluate them through an adversarial, economics-of-trading lens. Three findings summarize the evidence.

First, conditional diffusion models are competent P-measure generators. Static validation shows accurate means, volatilities, and overall distributional shape, and the P-model earns robust profits against a GBM-based market maker on European and Asian options, where

Table 16: P–Q game: snowball, KO 105%, KI 80%, coupon 15%

Spread Δ	0	0.5%	1%	1.5%	2%
Cumulative P&L	−3,714,555	−5,677,526	−7,571,108	−9,448,353	−11,337,305
Trades executed	821	798	776	766	763
Long positions	0	0	0	0	0
Short positions	821	798	776	766	763
Win rate	42.51%	42.11%	41.37%	40.99%	40.24%
Annualized Sharpe ratio	−0.90	−1.39	−1.88	−2.36	−2.84

terminal-value and path-average fidelity matter most.

Second, the model class and the training objective govern different statistics. Replacing Gaussian driving noise with α -stable noise—the DLPM of Shariatian et al. (2024) trained with its standard loss and no financial loss engineering—dominates both Gaussian variants on every conditional distribution-shape metric, halving the KS statistic and passing the KS test on average. The composite finance-aware loss, by contrast, is what recovers pooled tail mass for the Gaussian model (excess kurtosis 6.45 versus 3.19 for the simple loss), but it does so by inflating extremes indiscriminately, degrading per-window shape. The production model’s static validation carries the familiar Gaussian signature—a kurtosis difference of 3.56, by far its largest error—and the economic consequences surface exactly where extreme events dominate: lookbacks, accumulators, and deep-KI snowballs.

Third, the cross-family benchmark tempers any single-winner narrative at small training scale. A maximum-likelihood neural SDE is the best marginal calibrator but is structurally incapable of heavy tails or clustering; zero-shot Chronos attains the most realistic unconditional tail mass yet consumes no contract conditions and trails on conditional shape; the DLPM is the only conditional generator that combines near-best calibration with a noise mechanism capable of expressing tails, though realizing market-level kurtosis requires training beyond our matched ablation budget.

Fourth, and most consequentially, distributional fidelity converts into economic value exactly where theory predicts. Replaying the P–Q game with the matched-budget generators,

the Gaussian model reproduces its deep-knock-in failure mode—losing 1.1–2.3 million per million notional on snowballs while holding an almost uniformly one-sided book elsewhere—whereas the DLPM earns more than 8 million at every spread level on the same product with balanced positioning and win rates near 59%. On tail-dominated payoffs, the heavy-tailed generator is not merely statistically better; it is the difference between profit and loss.

Future work proceeds along four lines: training the DLPM to full convergence with an unbounded noise head, where its mechanism should translate into market-level kurtosis; risk-neutral calibration of the generator into a trainable Q-pricing model suitable for market making; adversarial co-training of the Q pricer and P trader; and extension to multi-underlying systems modeling cross-asset dependence, enabling valuation of rainbow and basket exotics.

Table 17: P–Q game under matched budgets: DLPM versus DDPM-complex

Greediness g	0	10%	20%	30%	40%
<i>European call ($K = S_0$)</i>					
DLPM P&L	66,109	60,723	52,644	40,371	34,025
DDPM-complex P&L	93,663	83,005	72,347	62,153	51,515
DLPM win rate	57.0%	57.0%	56.4%	55.2%	54.0%
DDPM-complex win rate	36.3%	35.7%	35.1%	34.4%	33.7%
<i>Asian call (arithmetic)</i>					
DLPM P&L	28,418	21,458	16,395	12,059	9,561
DDPM-complex P&L	50,716	44,636	38,555	32,474	26,393
DLPM win rate	51.0%	51.1%	50.1%	49.3%	48.6%
DDPM-complex win rate	35.9%	35.6%	34.8%	34.5%	34.0%
<i>Lookback call (fixed strike)</i>					
DLPM P&L	42,585	27,590	13,736	8,641	6,236
DDPM-complex P&L	181,934	163,748	145,562	127,376	109,190
DLPM win rate	56.5%	55.0%	53.2%	51.8%	50.4%
DDPM-complex win rate	49.7%	48.1%	46.8%	45.5%	43.5%
<i>Accumulator ($d = 80\%$, $KO 150\%$)</i>					
DLPM P&L	463	429	175	180	118
DDPM-complex P&L	1,524	1,252	1,002	593	-1
DLPM win rate	59.8%	60.2%	60.8%	61.7%	57.8%
DDPM-complex win rate	48.7%	45.7%	43.2%	39.1%	35.0%
<i>Snowball ($KO 105\%$, $KI 80\%$, 15%)</i>					
DLPM P&L	10,272,947	8,920,463	8,452,940	8,397,414	8,272,525
DDPM-complex P&L	-1,057,795	-2,259,836	-2,130,035	-1,633,041	-2,048,645
DLPM win rate	58.9%	59.2%	59.3%	59.3%	58.8%
DDPM-complex win rate	49.0%	48.4%	48.1%	47.8%	48.1%

Cumulative P&L and win rate of each P-model against the same GBM Monte Carlo market maker (512 simulations per window) over the 827 CSI 1000 test windows. Both P-models are the 4,000-step generators of the controlled ablation (Section 5); each values a contract by the discounted average payoff over 128 paths generated with 50-step accelerated sampling (DLIM for the DLPM, DDIM for the Gaussian model). Trades execute when the relative P–Q price gap exceeds 10%; snowball spreads are absolute ($\Delta = 0.05g$) and accumulator spreads scaled by $0.5g$, as in Section 6. Snowball P&L per 1 million notional.

A Network Architecture

Backbone. We use a one-dimensional conditional U-Net (Ronneberger et al., 2015) with base width 64, channel multipliers (1, 2, 4, 8), dropout 0.1, and 17.0 million parameters. The encoder consists of residual convolutional blocks and downsampling layers that halve sequence length while increasing feature dimension; skip connections preserve encoder features for the decoder, which restores resolution by upsampling and concatenation. Linear attention operates at each resolution and full self-attention at the bottleneck. Each residual block is

$$\text{ResBlock}(x) = x + \text{Conv}\left(\text{SiLU}(\text{Norm}(\text{Conv}(x)))\right), \quad (26)$$

with conditioning injected via feature-wise scale-shift modulation.

Time embedding. Diffusion steps enter through sinusoidal embeddings (Vaswani et al., 2017),

$$\text{TimeEmb}(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \dots, \sin(\omega_{d/2} t), \cos(\omega_{d/2} t)], \quad \omega_i = 10000^{-2i/d}. \quad (27)$$

Condition embedding. The two categorical identifiers pass through learned embedding tables (country: 64 dimensions; underlying index: 128 dimensions), the five numerical features through a linear projection with LayerNorm and SiLU (32 dimensions); the streams are fused by learned weighted concatenation and a two-layer MLP into a 128-dimensional condition vector,

$$\text{CondEmb}(c) = \text{MLP}\left([w_n \phi_n(c_{1:5}); w_c E_{\text{cty}}(c_6); w_x E_{\text{idx}}(c_7)]\right) \in \mathbb{R}^{128}, \quad (28)$$

which modulates every residual block together with the time embedding. The index embedding is deliberately the widest stream, as underlying identity carries the strongest style information.

B Training and Sampling Hyperparameters

Table 18: Hyperparameters of the controlled ablation

Diffusion steps T	500 (cosine schedule)
Stability index α (DLPM)	1.8
DLPM loss	unsquared L_2 , single-draw a_t , clamp $a_{\max} = 100$
Optimizer	Adam, learning rate 10^{-4} , cosine decay
Training steps / batch size	4,000 / 64
EMA decay	0.995
Composite-loss warm-up	15% of training steps
Return scale s_{vol}	0.09
Sequence length	252 (masked, variable tenor)
Sampling (evaluation)	full-chain ancestral, 500 steps, EMA weights
Sampling (accelerated)	DLIM / DDIM, 50 steps

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